# Soham Dadhich

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.41 |
| Tagore Public School, Vaishali Nagar, Jaipur | Class XII (CBSE ) | 2022 | 92.20% |
| Maheshwari Public School, Jawahar Nagar, Jaipur | Class X (CBSE) | 2020 | 90.20% |

## WORK EXPERIENCE

**Siemens AG | Technical Intern**

- Implemented **SEAL** (Self-Adapting LLMs, MIT 2025) from scratch, where a model generates its own fine-tuning data and reinforces only the edits that improve accuracy, **reproducing the paper's benchmark at 30.27% against its reported 37.0%**
- **Improved SEAL's reward design** and extended it to **code generation**, a domain the paper does not cover, taking non-compiling solutions from **18 to 0 of 50**. The redesign was filed as an invention disclosure, with details withheld
- **LoRA fine-tuned T5 / Flan-T5 embeddings** on OCR'd Siemens product manuals to fix retrieval that generic embeddings were failing, **cutting perplexity from 72 to 37** across an 8,448-chunk corpus
- Evaluated retrieval on 54 queries, **raising Recall@10 from 59% to 65%**, with the entire gain coming from hard multi-context queries rather than simple ones
- **Tech Stack:** Python, PyTorch, PEFT/LoRA, vLLM, AWS EC2

## SKILLS

- **Languages & Tools:** SQL, Python (pandas, NumPy, matplotlib), Excel, C++, Go
- **Analysis:** Experiment design, metric definition, segmentation analysis, benchmarking methodology, A/B testing
- **Statistics:** Probability, hypothesis testing, statistical significance, regression
- **ML / DL Fundamentals:** Classical ML, Deep Learning, Transformer architecture, NLP
- **CS Fundamentals:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

## PROJECTS

**Stock Price Prediction |** Stock_price_prediction

- Built and back-tested an **LSTM model** on historical price series, reaching **58% directional accuracy on held-out data**
- Ran **time-series, volatility and correlation analysis** to select features and verify the model was not fitting noise
- **Tech Stack:** Python, pandas, NumPy, TensorFlow, matplotlib, seaborn

**Key-Value Store |** Key-Value_Store

- Built a **networked key-value store from scratch in C++17** on POSIX system calls alone, with sharded storage, crash recovery, transactions, replication and networking
- Added a **write-ahead log so data survives a crash**, and batched commits to raise **write throughput 37x** without weakening the durability guarantee
- Built two interchangeable **server models, a thread per client and a single-threaded event loop** over non-blocking sockets with output buffering, plus a benchmark harness to compare them under load
- **Tech Stack:** C++17, POSIX, CMake, GoogleTest

## RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks
- **AI / ML:** Introduction to Artificial Intelligence, Computer Vision
- **Mathematics:** Calculus, Differential Equations, Elementary Linear Algebra
- **Applied Computing:** Fundamentals of Scientific Computing, Computational Methods in Materials Science

## SCHOLASTIC ACHIEVEMENTS

- **Filed an invention disclosure at Siemens AG** for a redesign of the SEAL self-learning method for code generation

## POSITIONS OF RESPONSIBILITY

**Hospitality and Logistics Coordinator, TEDx IIT Hyderabad**

- Mentored **20+ Assistant Managers** and ran guest coordination and logistics across multiple workshops with a **daily footfall of 600+**
- Managed catering and venue vendors for the event series

## EXTRACURRICULAR

- Won **Gold and Silver in swimming** at MILAN, the inter-hostel championship at IIT Hyderabad